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Policy advisory systems in times of crisis: challenges for expertise in policymaking

Abstract

Policy advisory systems which provide evidence-informed policy advice to governments are placed under great stress in times of crisis and rapid change. Crisis situations may give rise to conflicting interpretations by stakeholders about how to respond. These conflicts are compounded by inevitable gaps in knowledge and uncertainties about how the crisis conditions might develop. All governments face unruly challenges from time to time, although the contemporary era seems to be characterised by very frequent crises. These take many forms, ranging from financial and political disruptions through to health pandemics, natural disasters, and threats to national security.

Such decision-making stresses have been especially evident during the COVID-19 crisis, commencing in 2020. The pandemic crisis took different trajectories in various countries. This introductory article introduces a group of research studies that analyse various aspects of how governmental policy advisory systems responded to this crisis. Policy advice was formulated against a background of controversies about how to prioritise health outcomes, economic livelihoods, and social interaction. These research studies analyse policy advisory systems with particular attention to the quality of the available knowledge base, the disciplinary mix of expertise within advisory bodies, the roles of experts as either trusted insiders or as external commentators, as providers or 'shapers' of knowledge, and the degree of (in)formality in their relationships with politicians. These core issues are also addressed in research on creeping crises such as climate change. Taken together, the articles provide insights into how the knowledge provided through policy advisory systems is informing crisis governance.

Introduction – crisis and uncertainty

The aim of this special issue is to identify and describe how the recurrent crises and turbulence of recent years have impacted the policy advisory systems of modern governments, with particular reference to the COVID-19 pandemic. The case studies that follow demonstrate that the institutional capacities of governments, across a wide range of countries, have been challenged by crisis conditions and by the public controversies engendered by crisis situations. The core theme is the institutional relationships between expert advisors and political decision-makers, with a focus on how these relationships were questioned and modified under conditions of conflict and crisis. These debates have implications for broader issues such as democratic accountability and legitimacy, and effective and efficient governance, but these broad themes are not tackled here. In short, the special issue aims to advance the study of the relationship between experts in policy advisory systems and political decision-makers, under conditions of crisis. This introductory article provides an analysis of the key concepts and research themes examined in the case studies and concludes with a brief account of the contributions advanced by the case studies.

We are living in turbulent times. Governments have been confronted by multiple interacting crises, like the COVID-19 pandemic, global warming, and economic instability. Decision-makers are faced with complex challenges, where the problems are 'surprising, inconsistent, unpredictable, and uncertain' (Ansell, Sørensen and Torfing, 2021, p. 950). These unpredictable and sometimes contentious problems are likely to have large impacts – not only on social and economic wellbeing, but also on the capacities of institutions to anticipate and mobilise effective responses. The impacts of these crises on policy governance arrangements can be detected across diverse fields such as health, finance, safety and security. Similarly, in the private sector, business executives have been developing conceptual frames to make sense of living in a ‘VUCA’ world, characterised by ‘volatility, uncertainty, complexity, and ambiguity’ (Bennett and Lemoine, 2014).

Crises draw attention to governmental capacities for rapid action and adaptive responses, with a high premium on risk assessment and organisational coordination (Christensen et al, 2016). But behind the scenes, the quality of policy responses may partly depend on the analyses and recommendations provided by expert advisory bodies. Research on expert advisory bodies has examined the different ways in which government agencies have selected ‘relevant and legitimate’ forms of expertise, and how such knowledge has been incorporated in policy advisory systems (Craft and Wilder, 2017; Crowley and Head, 2017). Researchers have also raised the prospect that more collaborative, robust and adaptive policy advisory systems might become necessary for addressing future crisis situations (Zaki and Wayenberg, 2020; Boin and t’Hart 2022).

In recent years, the COVID-19 pandemic has sharpened the focus of academic research on the institutional capacities of crisis response and the potential roles of experts and knowledge partnerships. For example, health experts were heavily involved in providing advice and recommending strategies for coping with the pandemic, with messages directed not only towards government decision-makers but also to the general public (Boin, McConnell and ‘t Hart, 2021; Galanti and Saracino, 2021; Pattyn, Matthys and Hecke, 2021). Given the large number of recent international crises, such as financial meltdowns and pandemics, the research literature on crisis management has been reconsidering how various forms of expertise can be mobilised. Strategies that rely on a narrow band of specialists to address complex and fast-moving scenarios are increasingly seen as inadequate. Instead, there is growing support for using broad-based approaches, such as multi-stakeholder forums and knowledge partnerships, to deal with situations where no single group has sufficient knowledge and experience to provide authoritative advice and build public confidence (Lavazza and Farina, 2020; Moore and MacKenzie, 2020; Rajan et al., 2020).

Thus, according to recent work on collaborative approaches (Ansell, Sørensen and Torfing, 2021), both public and private actors will need to engage in ‘flexible adaptation’ and ‘proactive innovation’ to generate governance solutions for turbulent challenges. This perspective goes beyond promoting the goal of resilience, which generally focuses on the capacity of social groups, businesses and institutions to bounce back to their previous state in the aftermath of crises and challenges. While the resilience literature also notes the importance of individuals adapting in positive ways to enhance their pre-crisis functioning (such as increasing their skills and capacities – Laureys and Easton, 2019), the ‘robustness’ literature is more focused on the importance of building institutions and processes that can withstand shocks and disturbances (Capano and Woo, 2017; Howlett, Capano and Ramesh, 2018). For example:

> In turbulent situations, foresight, protection, and resilience are not enough. Instead, the public sector must meet turbulence with robust strategies, where creative and agile public organizations adapt to the emergence new disruptive problems by building networks and partnerships with the private sector and civil society. (Ansell, Sørensen and Torfing, 2021, p 952).

In the face of contemporary crises, effective communication among epistemic communities in different sectors is all the more valuable. The important inter-dependencies between the sectors – government, private business, community organisations, academia, media and the (natural) environment – need to be recognised and strengthened. Productive interaction across the sectors has been well documented in knowledge-intensive industries, as exemplified in the Quintuple Helix framework (Carayannis and Campbell, 2010) for encouraging creative and innovative solutions for societal problems. The information and communication aspects of collaboration, for crisis management more broadly, are increasingly dependent on the quality of the digital (e-government) dimension of modern governance (Mergel et al, 2018).

The quality of policy advisory systems influences the capacity of governments to manage both mainstream issues and unexpected major crises. Advisory systems have normally been designed for providing expert information about known problems, and for recommending policy measures or programs that can stabilise or improve the situation (Jensen et al., 2022, pp 223-4). However, under conditions of crisis and disaster, there are likely to be serious gaps in knowledge, disagreement about the scope of the risks, and conflict about appropriate measures to contain the negative impacts (Drennan, McConnell and Stark, 2015). The future design of advisory systems may therefore need to adapt to these unpredictable and fast-moving situations by considering more inclusive and flexible forms of engagement with experts and stakeholders (Boin and Lodge, 2016).

In liberal democracies, the traditional narrative has been that various forms of expertise are valued for their contribution to the quality of political debate and governmental decision-making. Elected government ministers hold executive authority to make decisions, but they are usually expected to take account of expert advice. To the extent that experts are seen as objective, reliable and credible sources of information and advice, they can provide a further layer of legitimacy for the policy process owing to their specialised knowledge. However, in the contemporary era of turbulent and politicised crises, the idealised picture of experts as objective and non-partisan is less persuasive to the public, owing to several factors such as the interdependency of ministerial offices and policy advisory systems; the partisan mediatisation of crisis issues leading to increased pressure on decision-making; and the very rapid dissemination of information (and disinformation) through digital channels of communication.

Among OECD liberal democracies, the institutional relationships between expert knowledge systems and governmental processes are quite diverse (OECD 2015; Hustedt and Veit, 2017). Much more diversity is evident when the scope of research is extended to include the full range of Low and Middle-Income Countries (LMIC) and non-liberal regimes. Leaving aside the likely effect of autocratic leadership styles, several key factors may seriously limit the capacity of governments to deploy the most effective available responses. These factors include resource issues (e.g. gaps in data collection and analysis, lack of science advisory infrastructure, limited funding for research and evaluation), and network issues (e.g. poor linkages between academia and policymaking). Recent research indicates that decision-makers in LMICs prefer using evidence obtained from similar settings or regional contexts, rather than deferring to OECD-centred research (Damba et al, 2022; Witter et al, 2019). Research programs have not yet yielded significant work on the institutional structures and operational mechanisms of science advice processes in LMICs, taking into account the socio-economic constraints, socio-cultural norms and types of political leadership in those countries.

Studies of policy advisory bodies in liberal democracies tend to show that experts are often sharing the 'driving seat' during crises, but that high-level policymakers decide on the key strategic directions which entail difficult choices and trade-offs among competing goals and stakeholder interests. Thus, the research themes of this special issue draw upon three interrelated bodies of literature which inform and shape the landscape of expert-policymaker relations under conditions of crisis and turbulence. The core focus is to examine the various contributions of experts, inside and outside government, to policymaking on complex issues under conditions of conflict and uncertainty. A related concern is how these issues become highly politicised, with implications for the (perceived) politicisation of expert advice. Finally, it is important to frame this analysis in the context of crisis governance. To set the scene, it is necessary to clarify different types of crises (recurrent, creeping, and acute) before proceeding to our main focus on expert policy advisory processes.

The crisis literature has identified several major forms and features. Crises are understood as serious forms of disruption, characterised by threat, urgency, and uncertainty. The term ‘crisis’ generally denotes a perceived major threat – social, political, financial, etc – to ‘core values or functionality’ of a national or an organisational system, ‘which must be dealt with under conditions of uncertainty’ (Boin et al. 2018, p. 24). There is some overlap with the fields of disaster management and emergency response, but the latter are usually about responding to physical hazards such as industrial explosions, storms, floods, droughts, wildfires, earthquakes, and volcanic eruptions. The terminology of ‘disasters’ and ‘emergencies’ may also have more specific meanings within legislative or regulatory instruments, but we will use the generic term ‘crises’ in this article.

The literature distinguishes between various types of crises. For example, crises unfold differently according to variations in policy sectors, spatial scales, and developmental timelines. Thus, a crisis might develop within a specific sector or policy domain, such as finance, health, transportation, housing, energy, or information technology; but spillover effects could also greatly affect related policy domains, and some of these threats and responses are likely to recur from time to time. A specific crisis might have only local impacts within a country, whereas other crises might cascade across international borders. For example, civil wars and famines may generate waves of refugees, who are displaced from their homelands and who seek security elsewhere in unfamiliar countries. Thus, crises may interact with each other in a cascading and compounding manner.

Finally, while most of the crisis literature has focused on ‘acute’ crises which emerge rapidly and require immediate responses, other types of crises are slow to develop over many years (Boin, Ekengren and Rhinard, 2021). ‘Creeping’ crises, such as climate change, may fail to achieve the public salience necessary for attracting political attention and policy debate. In the absence of widespread awareness of emergent risks, these slow-onset crises may escape priority concern until urgent symptoms of harm have been recognised by political and media opinion leaders (Boin, Ekengren and Rhinard, 2020; Staupe-Delgado and Rubin, 2022).

Dimensions of expertise in policy and politics

In policy and politics, the contributions of experts have always been important, but the form and content of valued expertise is highly variable. In liberal democracies since the 1950s, there has been a broad consensus that public policy and public administration should be substantially anchored in empirical and scientific knowledge about trends and issues (Adams 2004, pp 30-31). The sources of expert-based information include the social and natural sciences, policy analytics, governmental reports, and research coming from universities, think tanks, and consulting firms. Policy-related experts include policy analysts, scientists, and researchers in governmental and non-government organisations. There is much diversity across these groups of experts in terms of their financial incentives and professional cultures. Many have careers that include working in more than one sector, such as academia, nonprofit, private corporations, think tanks and government (Weible 2008, p 216).

The research literature about ‘expertise’ is very broad, multi-disciplinary, and draws attention to several criteria for recognising expertise including formal qualifications and peer reputation (Eyal 2019, ch 2). Much of the recent literature draws a distinction between the term ‘technical specialists’ whose knowledge scope is specific and narrow, and the term ‘experts’ whose deeper and wider knowledge has developed through experience in different situations (Goldman 2001; Grundmann 2018). However, for the present purpose of examining public policy advisory systems, we take a pragmatic view of experts as those employed in professional roles in the policy-related functions of government, together with those employed in non-government organisations (such as research institutes, think tanks, consulting firms, and business or community organisations) who have been invited to be members of governmental policy advisory bodies. The valued forms of knowledge might range from scientific research through to knowledge of policy options, policy instruments, program management, and processes for building capability and support.

The ongoing demarcation between expert scientific knowledge and the lay knowledge based on experience has had the effect of conferring scientific experts with 'epistemic authority' (Gieryn 1983). It was notable, in the first year of COVID-19, that government leaders relied heavily on the authority of health scientists to defend unpopular restrictions and lockdowns (MacAulay, et al, 2023; Lynggaard et al, 2023). However, such reliance on the epistemic authority of health scientists was short-lived as attention turned to economic recovery and social connectivity. The temporary elevation of health science as the key to public safety did not shift the long-term underlying distrust of expertise in many countries where anti-elitist political rhetoric has been amplified by populist media channels (Nichols 2017). Debate about the threats of technocracy and elitism has a long history, not only among populists. Many writers on democracy and accountability have championed more participatory forms of decision-making, voicing concerns about the scientization of politics and the need for open discussion of complex policy issues (Fischer 1990). Technical specialists may have superior knowledge about technical processes, but the policy choices for complex issues often turn on normative considerations such as social values and the distributional outcomes for a diverse range of stakeholders (Lavazza and Farina, 2020, p 10).

The roles of experts are likely to vary in accordance with different policy domains, and the institutional patterns in various policy subsystems and advisory mechanisms. For example, researchers have drawn distinctions between unitary, collaborative, and adversarial subsystems (Weible, Pattison and Sabatier, 2010). The relationships among experts might exhibit a high degree of agreement or consensus; in other cases, the experts may be engaged in attempts to reconcile differences to achieve shared goals and approaches; and in other fields the disagreements are so fundamental that the policy domain is characterised by polarisation and adversarial positions. Among scholarly experts, these matters of dispute might include relevant theories, methods, empirical data, and standards of validation.

In OECD countries, but also increasingly in other countries, policymakers seek expert advice through scientific advisory committees (SACs) not only on technical issues but also on complex social issues. A recent scoping review of scientific advisory committees in Europe and the United States (Groux, Hoffman and Ottersen, 2022) proposed six broad analytical categories for distinguishing between the types and functions of SACs. Their typology was based on distinguishing between the policy sector in which the SAC was established (e.g., national security, health or education); its jurisdictional level (e.g. national, or sub-national); the duration or longevity of the SAC (e.g. ad hoc, or permanent); the target audience for its advisory outputs (e.g. Ministers, or broader networks); its degree of independence (arms-length or embedded); and the type of advice it typically delivers (e.g. explanatory or prescriptive recommendations). Each of these dimensions is important for understanding the operating landscape of each SAC, and case studies that explore these dimensions could throw light on which institutional designs could make SACs more effective in specific contexts. Given the OECD-related background of these studies, much more attention is required to provide a comparative international picture of SAC developments in LMICs.

In the following sections, we very briefly address several analytical themes in the research literature on the dynamics of expert policy advisory systems. In particular, we explore the positioning of experts as insiders or outsiders in the political system, the formal and informal dimensions of their advisory roles, their influence as knowledge providers or shapers, and their either non-partisan or politicised relationships to the political system. We will then consolidate the discussion by focusing on the relations between experts and policymakers during crises such as the pandemic.

Insider/outsider positioning

The proximity of experts to decision-makers is a key element in how a policy advisory system works. The relevant experts can be either within the public bureaucracy or based in diverse non-government organisations. A century ago, Weber's claim about the growing power of bureaucrats (Weber 1958, pp 228-235) was based on his concern that ministers would be over-reliant on the deep content knowledge of their bureaucratic advisors. The modern situation is different, insofar as the sources of advice for policymakers have proliferated, and many non-governmental stakeholders are regularly involved in policy discussions with ministers (Craft and Howlett, 2013). Ministers have considerable discretion concerning where they seek advice and which sources are seen as trustworthy, so the insider status of policy bureaucrats has been weakened in a more pluralist advisory context.

Studies of business lobbies and pressure groups in recent decades have shown not only that public bureaucrats have lost their monopoly over ministerial access, but also that some external lobbyists have been very influential. In differentiating between insiders and outsiders, Page (1999) argued that insiders were those who had frequent contacts with senior policymakers, and who were regularly consulted on all relevant matters in their field. Hence, they could influence policy through direct contacts. By contrast, outsiders are those excluded from the corridors of power; hence, they were more likely to engage in alternative methods to influence policy, such as organising demonstrations and running critical commentaries in the media. He suggested that there is a spectrum of access to decision-making processes, so the binary insider-outside labels can be misleading (Page 1999, p 212). In another study, examining the policy influence of bureaucratic experts, Page argued that it is the *status* or reputation of expert groups, committees, and institutions, rather than the specific *content* of their expertise, that mainly accounts for their influence (Page 2010, p 270).

This analysis underlines the importance of the proximity of experts to decision-makers as a key element within an advisory system. The literature also examines the design aspects of policy advisory systems (Aubin and Brans, 2021) such as the scope and powers of advisory bodies, and the basis of making appointments (such as specific knowledge expertise, or the representation of sectoral interests).

Formal/informal dimension

The interaction between formal organisational structures and informal social relations is one of the basic issues in organisational theory. It is well accepted that formal and informal ties interact and overlap within organisations (Yakubovich and Burg, 2019). While difficult to research, an interesting dimension is the degree of formality and informality in the relationships and interactions between experts and politicians. For example, research on lobbying in relation to climate policy (Coen 2007; Gullberg 2008) showed that expert-politician relations can be based predominantly on political or personal networks (individual connections between scientists and policymakers, ad hoc uptake of scientific research, conferences, and workshops) rather than simply relying on formal public administrative processes.

Informal relations are built up within and between groups of experts and politicians. These links can take a variety of formats depending on people's background, ranging from 'associations for alumni' to trusted relationships and friendships that transcend occupations and status. Again, this dimension is not binary and relationships between people can have both formal and informal dimensions. This adds complexity to the relationships found within expert advisory structures and between involved experts and politicians.

Experts as providers or shapers

The research literature on experts in the policy process often raises the question of experts as knowledge 'shapers' or as knowledge providers. A traditional view of technical expertise stressed that objectivity and detachment are desirable (Pielke 2007). Many experts, including within the health sector, are content simply to provide scientific information rather than consider the policy implications. In a recent UK-based interview study (N=93), most scientists expressed a desire to provide evidence for informing policymakers, rather than a desire to advocate for specific policy options (Atkinson et al 2022). Another recent study, based on interviewing health scientists on advisory boards in Europe, showed a similar pattern of uncertainty about the scope of their roles, including how their health expertise might contribute to addressing broader inter-disciplinary policy problems (Colman, Wanat et al, 2021).

However, others have argued that experts should adopt a more active role by seeking to become knowledge-shapers rather than just being information-providers. In this approach, experts are also engaging actively with others to solve problems by framing and communicating the nature of the problem and pathways for improvement (Hordijk and Baud 2006; Sprujit et al. 2014). In policy studies theory, such as the advocacy coalition framework, experts are seen as brokers and entrepreneurs in terms of policy debate and policymaking (Weible et al, 2020; Petridou and Mintrom, 2021). In the climate-change literature, the role of experts has been seen as not only providing scientific information, but also engaging strategically with the public and policymakers to enhance mobilisation and to shape key narratives and goals regarding climate change (Howarth et al. 2020; Biermann 2021; Livingston 2023).

Experts take up different roles in policymaking depending on the nature of the issue and the prominence of expert knowledge in framing the issue. In recent years, there has been strong involvement by non-government experts in climate change policy development through linking science, politics, and advocacy (Pralle 2009; Esguerra et al, 2017). In the traditional agenda-setting literature, experts in policy communities were characterised as primarily disseminating evidence-based information to policymakers and the public (Kingdon 1995, p 228). However, experts who take on wider roles as public knowledge-shapers contribute importantly to the framing of issues, both in the sphere of formal decison-making and more broadly among the general public. Some experts play important public roles in media advocacy, agenda-setting, and policy debate (Dorfman and Krasnow 201; Christensen 2021). Expert advisors are often encouraged by the media and by political decision-makers to participate in the public debate, in order to clarify issues, increase public understanding, and build legitimacy around new policies.

Politicisation: inevitable or useful

Politicisation is a term with several different meanings. It can be identified with the conflictual aspects of partisan political debate; or it can be focused on cases of in-group patronage, favouritism, and exclusion; or it can be seen more positively as a necessary quality for attracting priority support by policy entrepreneurs and political leaders. Politicisation is sometimes contrasted with the notion that liberal democracies should accord the highest importance to knowledge and expertise in public policymaking and public debate. Indeed, there has been a broad push towards 'evidence-based' or 'evidence-informed' policymaking since the 1990s (Head, 2016; Christensen 2021), entailing a greater reliance on scientific research expertise. With a growing emphasis on expert knowledge and expert judgement comes a need to ensure supplies of reliable and relevant information. The production and quality assurance of knowledge for specific policies and regulations requires careful oversight (Majcen 2017). Experts may hope they are contributing to an objective policy process based on scientific expertise, evidence, facts, and rationality. However, their endeavours are not generally used in a value-free way by decision-makers, who are more inclined to seek evidence-informed arguments in support of their chosen options (Cairney 2016).

Research on the politicisation of policy advisory systems draws attention to several common practices such as politically appointed ministerial advisors, partisan selection of advisory committee members, and biased framing of the terms of reference for public inquiries and advisory bodies (Craft and Halligan, 2017; Fischer, Wenholt, Rowe and Frewer, 2014; Hustedt and Veit, 2017). The politicisation of an issue may make it more challenging for scientific expertise to set the terms of discussion in policymaking (Radaelli 1999; Weingart 1999; Lacey et al., 2018). With complex issues, such as climate change, policymaker interests vary and scientific expertise may be reframed for political ends (Liberatore and Funtowicz, 2003; Head 2016, p 474). When issues are heavily politicised, the distinct and separate roles of expert scientific advisors, experienced policy bureaucrats, and ministerial decision-makers and their staff, can become blurred (Hodges et al, 2022; Christensen and Lægreid, 2022). In these circumstances partisan expertise may become more likely to over-ride the professional advice of public servants and expert committees. Jasanoff (2003) notes that government decision-makers may move between two contradictory positions: either they assert that all expert knowledge is 'biased' – thereby justifying their politicised appointments to advisory bodies – or alternatively they adopt 'the elitist view of expertise as superior knowledge'. In order to overcome or reframe this dichotomy, she argues that expert advice should itself be subjected to close scrutiny especially when it has consequences for regulatory decision-making:

> experts exercise a form of delegated authority and should thus be held to norms of transparency and deliberative adequacy that are central to democratic governance (Jasanoff 2003, p 157).

Expert-politician relations during crisis

The role of experts in policy advisory positions varies across policy fields and policy issues, depending on institutional contexts and policy history including knowledge uncertainties and partisan disagreements. This brings us to the core questions raised in this special issue: How do conditions of crisis and turbulence affect the ways in which policy advisory systems function? How do crisis conditions affect the activities of public agencies in gathering and appraising reliable information, consulting with key stakeholders and the public, assessing future risks, and developing sound policy options for the consideration of ministers?

Recent research shows how crisis management obscures the classic dichotomy between politics and administration, including the provision of expert advice. Research on expert involvement in policymaking during diverse types of crises is extensive (Bækkeskov 2016; Topp et al., 2018; Cairney and Oliver, 2020; Boin, Ekengren and Rhinard, 2021; Laage-Thomsen and Frandsen, 2022). In a context of crisis or high turbulence, the pre-existing institutional arrangements for policy advice may either facilitate or frustrate appropriate involvement of experts. In response to crisis situations, political systems are usually expected to utilise existing practices (path dependence) even if the threats, urgency, and uncertainty create pressures and opportunities for new approaches to crisis management. On the other hand, during times of turbulence the existing advisory structures can be adapted, modified, or even circumvented depending on the type of crisis (Boin and Lodge, 2016; Boin, McConnell and ‘t Hart, 2021). The dynamics of crisis decision-making are likely to expose weaknesses in resources, analytical expertise, and capacity to adjust rapidly to changing circumstances. Below we briefly highlight some of the key factors influencing the performance of policy advisory systems under the pressure of crisis conditions (fast-spaced expertise-informed decision-making; diversity of expertise; stakeholder agreement; politicization pressure; and the salience of expertise).

Fast-spaced expertise-informed decision-making

One factor influencing policy advisory systems is the need for fast-paced decision-making. Governments typically involve and rely on expertise when information and knowledge are scarce (Fakhruddin, 2020; Hodges et al. 2022). This has become increasingly required even in 'normal' times but is greatly intensified in crisis conditions (Boin et al., 2021; Comfort et al., 2020). Politicians need experts and expertise during crisis, as governments often lack sufficient, reliable information at a time when reluctance to act is not an option (Rubin et al., 2021). Ministers demand real-time information about the evolving crisis, which in turn puts pressure on existing channels of information-gathering and analysis. In an acute crisis there is no time for the rigorous and comprehensive 'long-term' science that experts usually prefer, and little time to double-check findings, or use extensive peer review, or commission new research. There are likely to be big gaps in the data, major difficulties in estimating the trajectory of the crisis, and large uncertainties about effective interventions to manage the situation. Instead, experts and decision-makers must rely on 'real-time' science or 'good enough' knowledge. Expertise must be exercised under unfavourable conditions such as lack of time (short deadlines), insufficient data, uncertain parameters for modelling, lack of skilled personnel, and lack of key supplies and facilities and support for vulnerable and disadvantaged groups.

Experts appear to be especially dominant in the first phase of a crisis, when there is an immediate need for person-linked trusted expert advice to make rapid responses to an uncertain and threatening situation (Hadorn et al., 2022). In this phase policymakers are often relying on trusted individual experts who already have a history of interaction with policymakers. This leads to better adoption of science-based information and advice than a system based on analysis of a complex evidence base. According to a comparative study of experience in Germany, Switzerland and Italy, countries with policy advisory systems designed to use personal expertise may be better placed to incorporate scientific knowledge into their decisions in times of acute crisis than are countries with policy advisory systems that relied primarily on research evidence and public reports (Hadorn et al, 2022).

Diversity of expertise

A second factor affecting policy advisory systems during crises is the choice and range of expertise that is drawn upon by executive government. Governments need experts to gain and maintain public sensemaking around crisis management, especially during the first phases of a crisis. The framing of a crisis in a particular way will influence the breadth of expertise that is seen as most relevant – for example, a national security crisis will accord priority to military and policing expertise (Christensen et al, 2013), whereas a deadly infection among livestock will accord priority to health and agriculture expertise (Frewer and Salter, 2002). The more ‘technical’ the problem, the more likely that external scientific expertise will be sought by policymakers. With complex issues, such as climate change, policymakers’ interests will vary and multidisciplinary scientific expertise may be more or less welcome (Liberatore and Funtowicz, 2003; Radaelli, 1999; Weingart, 1999). Some governments in liberal-democratic societies have begun to facilitate diversity within expert advisory systems in the belief that it stimulates scientific clarification and public understanding.

Stakeholder agreement

A third factor that can complicate the development of appropriate policy responses is the level of stakeholder agreement about values and goals. If stakeholders have different views about core principles and objectives that need to be observed during the crisis, the capacity of a democratic government to develop legitimate and effective response measures is further tested. Governments need experts to justify unpopular crisis interventions. Having good science is useful for decision-makers, but science itself does not resolve the difficult choices and trade-offs between competing perspectives. Stakeholder involvement and agreement are important here. It is democratically risky for politicians to make blunt interventions, based on advice that has not been scrutinised through scientific review and public debate (Rubin et al., 2021).

Politicisation pressure

A fourth factor affecting policy advisory systems during crises is politicisation. Some governments may amplify the perception of threat to justify actions that would not otherwise be accepted, whereas others may downplay the threat to avoid taking drastic actions (Boin et al. 2018, p 25). The prospect of growing politicisation of policy choices under crisis conditions has been discussed by several scholars (e.g. Christensen and Lægreid, 2022; Craft and Howlett, 2013). There is also a risk that reliance on rigorous evidence will be undermined when preferred 'insiders' occupy the key advisory roles (Rubin et al., 2021). Politicisation affects the tension between maintaining the independent character of expert scientific advice and the harnessing of that advice to political objectives. As noted previously, crisis pressures tend to blur the classic dichotomy between politics and administration, including how advice is accessed and managed (Christensen and Lægreid 2022; Weingart 1999).

Salience of expertise

A fifth factor is the salience of expert knowledge during crises, when threat, urgency and uncertainty emerges. Governments often give prominence to experts (also described as frontstage legitimation) in crisis situations (Hadorn 2022; Hodges et al, 2022). Increasingly, expert advisors – from inside and outside government – play a public communication role during crisis, being thrust into the media spotlight to support political leaders and provide public explanations and justifications of policy measures (Poulsen 2022). In general, the media play a large role in positioning experts in crisis situations, especially in the early stages of the crisis. However, the public salience of expertise can also be manifested in critical commentaries on crisis responses by experts who feel impelled to exercise their independent judgement. Giving visibility to experts may also be important from the viewpoint of democratic accountability and transparency, since unelected experts may be influencing decisions that affect citizens' liberties and livelihoods.

The above features of expert advisory systems in acute crises can be different from the dynamics found in slow-onset or creeping crisis, like climate change. Though urgency and uncertainty are major challenges, the main political challenge is agenda-setting or making the issues more salient in public discussion. Furthermore, it can be difficult to maintain public attention and momentum in relation to creeping crises, because interventions to curb these crises often need to be implemented over decades (Boin, Ekengren and Rhinard, 2020; Staupe-Delgado and Rubin, 2022).

Contributions in this special issue

The special issue includes the following contributions (see table 1):

(Table 1)

Easton, Yarnold, Vervaenen, De Paepe & Head describe expert perspectives on the changing relational dynamics of policy advisory systems during the first two years of the COVID-19 crisis in Belgium and Australia. Although both countries had policy advisory structures in place prior to the crisis; earlier research shows the different nature of the policy advisory systems in both countries during the pandemic. While Australia’s mostly *adapted* policy advisory structure remained relatively stable, Belgium’s *activated* and *ad hoc* policy advisory structures underwent ongoing changes throughout the crisis (Easton et al., 2022). Taking these findings into account; a qualitative analysis of 34 interviews with key expert informants is used to examine how expert advice was sought, provided, and utilized in crisis decision making; the specialized knowledge base of experts in relation to the goals of COVID-19 advisory bodies; and the relationships with other experts and policymakers. Their data, on the first two years of the crisis, shows the expanding breadth of perceived ‘relevant’ expertise as the scope of the challenge was re-framed from a health crisis to an all-encompassing socio-economic crisis. The experts interviewed illustrate the salience of expertise during crisis, the fast-spaced expertise-informed decision-making that took place and their experience with politicization pressure. Insights are given on how formal and informal dynamics unfold. Furthermore, this piece of research highlights that experts in advisory bodies are embedded in networks that facilitate links between different institutions, issue domains, and varieties of expertise. Therefore, effective crisis advisory bodies should incorporate experts from diverse backgrounds who can act as boundary spanners to link internal and external networks.

In “The promise and performance of data ecosystems”, **Howard & Hyland-Wood** present a case study of infectious disease modelling as part of Australia’s COVID-19 response. The authors suggest that the rise of ecosystem thinking in public policy and administration represents a new, radically decentred model of policy action. It focuses on ecosystems of data-driven modelling and experts who utilise government public health data. They present a conceptual framework that suggests effective data ecosystems undertake organic adaptation, display institutional leadership, and respond instrumentally to policymakers’ information needs. Their work is based on a case study of how Australia's infectious disease data ecosystem performed before and during the COVID-19 pandemic. In-depth interviews with data experts involved in infectious disease modelling in Australia revealed trade-offs within and between organic, institutional, and instrumental dimensions, which COVID-19 exacerbated. The authors conclude that data ecosystems represent a hybrid form of policymaking in practice and that tensions must be managed to maximise the contributions of data ecosystems for evidence-informed decision-making. They touch upon a crucial element in how to prepare better for crises in the future.

In 'Between naked power and technocracy', **Aagaard, Poulsen & Chatzopoulou** compare relations between experts and politicians during the COVID-19 crisis in Denmark, Greece, and the United States. For their comparative case study, data is collected from media, newspapers, and public documents. Their empirical findings show a high degree of national path dependency in national institutional arrangements in crisis management, connected with variations in increased pressure for politicisation. The authors indicate how systemic and governance characteristics affect expert politics responses in the three cases. The findings demonstrate a high resemblance in responses between Greece and Denmark, compared to the United States. Lastly, the authors formulate new research propositions for expert–politician relations during crisis situations that address issues of experts' roles in public sensemaking during crises, issues which are especially important in an era of recurring and overlapping crises.

Lynggaard, Exadaktylos, Jensen & Kluth explore the role of expertise in shaping policy responses during politicized/depoliticized processes. Drawing on unique empirical evidence from a survey among political science experts across 31 countries in Europe; they examine knowledge exchanges before, and at different stages within, the COVID-19 pandemic. They found patterns of change between normal policymaking conditions to conditions of politicization/depoliticization as triggered by the pandemic. It allows them to offer a novel typology, classifying the role of experts as leading, antagonistic, managerial, and auxiliary. During the Covid-19 pandemic a dramatic shift of the role experts occurred whereby even in countries where experts previously had an auxiliary role, leading decisions became the norm. They conclude that the role of experts oscillates depending on the issue at hand and the way that either political actors utilize expertise to legitimize decisions or experts themselves turn into political actors within politicized decision environments.

An important contribution to broaden insights on expert advisory systems in low- and middle-income countries (a gap mentioned above by Groux et al. (2022) is the research of **Naomi, Allen, Rima Mbaabu, Mendes, Ndongwe, Djindan, Gardia de Paredes & Kandanamull**. Their article on 'Evidence-to-policy pathways and pandemic governance' addresses comparative institutional insights from COVID-19 responses in low- & middle-income countries in the global South. Drawing on six individual case studies (Panama, Jamaica, Sri Lanka, Indonesia, Kenya, and Zimbabwe) representing Asia, Latin America and the Caribbean and Africa; the authors' highlight issues of weak science advice infrastructure, poor linkages between academia and policy, as well as limited research funding opportunities. At the same time their findings suggest that the presence of well-established institutions and pathways for science advice generally improved the uptake of scientific evidence in pandemic decision-making. Their analysis shows how contextual factors can severely impede processes of evidence generation and evidence-informed policy development. The author's point out that there exists a critical need for models of science advice adapted for local contexts to serve governments more effectively in bridging the evidence to policy gap.

Most studies have identified the main challenges faced by expert agencies involved in evidence-based policymaking as managing uncertainty, time pressure, and communication. However, less focus has been devoted to analysing the concrete challenges faced by expert agencies during creeping crises. Creeping crises are characterized by spatial and temporal fragmentation and elusiveness, which create an additional challenge for expert agencies: placing the crises on the political agenda. **Zaman, Rubin and Delgado** fill this gap with their contribution on "The scope for expert-led agenda setting during creeping crises: the curse of complacency". Comparing two global creeping crises: climate change (CC) and antimicrobial resistance (AMR), this article highlights two distinct strategies for influencing policymaking. The analysis shows how two expert agencies, the World Health Organization (WHO) and the United Nations Framework Convention on Climate Change (UNFCCC), pursue different strategies when setting the global agenda and influencing policymaking. The findings show that the WHO's approach to policymaking regarding AMR has been mostly guided by top-down, science-led, formal engagements and strategies. This approach has successfully increased the salience of the global challenge of AMR, providing strong, evidence based solutions, but it has been less successful in promoting the challenge onto the global political agenda. In contrast, the UNFCCC's approach to policymaking has relied more on horizontal, bottom-up, multidisciplinary, informal strategies. This approach has enabled a broader coalition of advocacy actors and placed CC persistently on the global political agenda. In this way, the article enhances our understanding of the role experts play in drawing attention to creeping crises.

Further insights on climate policy in the EU is provided by **Dupont, Rosamond & Zaki**. In "Investigating the scientific knowledge-policy interface in EU climate policy", the authors provide an historical overview of the evolution of knowledge exchange architecture for climate policy in the EU. They investigate whether evolutions in the knowledge architecture reflect shifts in the politicisation of climate change and focus on knowledge exchange with the European Commission. Drawing on 34 semi-structured interviews carried out between 2010 and 2022, their analysis reveals reveal connections between the development of the formal and informal aspects of the knowledge exchange architecture and the shifts in politicisation in different time periods. They find that when the politicisation of climate change led to a negative or constraining context, the informal aspects of the knowledge exchange architecture closed, making it more challenging for multidisciplinary scientific knowledge to enter the process. However, the formal aspects of the knowledge exchange architecture remained in place, even under constraining conditions. With their contribution, the authors provide a nuanced assessment of the connections between the effects of politicisation and the potential for meaningful scientific-policy knowledge exchange, contributing to literature on both the politicisation of climate change and on knowledge exchange architectures.

References

Adams, D. (2004). Usable knowledge in public policy. *Australian Journal of Public Administration*, 63(1), 29-42.

Ansell, C., Sørensen, E. and Torfing, J. (2021). The COVID-19 pandemic as a game changer for public administration and leadership? The need for robust governance responses to turbulent problems, *Public Management Review*, 23(7), 949-960.

Aubin, D., and Brans, M. 2021. Styles of policy advice, in M. Howlett and J. Tosun (eds), *Routledge handbook of policy styles*, London: Routledge, pp. 286-299.

Bennett, N., and Lemoine, G.J. (2014). What a difference a word makes – understanding threats to performance in a VUCA world. *Business Horizons* 57(3), 311-317.

Biermann, F. (2021). The future of 'environmental' policy in the Anthropocene: time for a paradigm shift, *Environmental Politics*, 30 (1-2), 61-80.

Boin, A., Ekengren, M. and Rhinard, M. (2020). Hiding in plain sight: conceptualizing the creeping crisis, *Risk, Hazards and Crisis in Public Policy*, 11(2), 116-138.

Boin, A., Ekengren, M. and Rhinard, M. (Editors) (2021) *Understanding the Creeping Crisis*. Palgrave Macmillan: Cham.

Boin, A., Hart, P.‘t. and S. Kuipers (2018). The crisis approach, in H. Rodríguez, W. Donner and J. Trainor (eds.), *Handbook of Disaster Research*, 2nd edition, Springer, pp 23-38.

Boin, A., and ‘t Hart, P. (2022). From crisis to reform? Exploring three post-COVID pathways. *Policy and Society*, 41(1), 13-24.

Boin, A., and Lodge, M. (2016). Designing resilient institutions for transboundary crisis management: A time for public administration. *Public Administration*, 94(2), 289-298.

Boin, A., McConnell, A., and ‘t Hart, P. (2021). *Governing the pandemic: A crisis management perspective*. London: Palgrave.

Cairney, P., and Wellstead, A. (2021). COVID-19: effective policymaking depends on trust in experts, politicians, and the public, *Policy Design and Practice*, 4(1), 1-14.

Capano, G., and Woo, J. (2017). Resilience and robustness in policy design: a critical appraisal, *Policy Sciences* 50(3), 399-426.

Carayannis, E., and Campbell, D. (2010). Triple helix, quadruple helix and quintuple helix. *International Journal of Social Ecology and Sustainable Development*, 1(1), 41-69.

Christensen, T., and Lægreid, P. (2022). Scientization under pressure – the problematic role of expert bodies during the handling of the COVID-19 pandemic. *Public Organization Review*, 22(2), 291-307.

Christensen, T., Danielsen, O., Lægreid, P., and Rykkja, L. (2016). Comparing coordination structures for crisis management in six countries. *Public Administration*, 94(2), 316-332.

Coen, D. (2007). Empirical and theoretical studies in EU lobbying. *Journal of European Public Policy*, 14(3), 333-345.

Comfort, L., Kapucu, N., Ko, K., Menoni, S., and Siciliano, M. (2020). Crisis decision-making on a global scale: transition from cognition to collective action under threat of COVID-19. *Public Administration Review*, 80(4), 616–622.

Craft, J., and Halligan, J. (2017). Assessing 30 years of policy advisory system scholarship. *Policy Sciences*, 50(1), 47-62.

Craft, J., and Howlett, M. (2013). The dual dynamics of policy advisory systems: the impact of externalization and politicization on policy advice. *Policy and Society*, 32(3), 187-197.

Craft, J., and Wilder, M. (2017). Catching a second wave: context and compatibility in advisory system dynamics. *Policy Studies Journal*, 45(1), 215-239.

Crowley, K. and Head, B.W. (2017). Expert advisory bodies in the policy system, in M. Brans, I. Geva-May and M. Howlett (eds), *Routledge handbook of comparative policy analysis*, London: Routledge, pp. 181-198.

Damba, F., Mtshali, N., and Chimbari, M. (2022). Barriers and facilitators of translating health research findings into policy in sub-Saharan Africa: a scoping review. *Humanities and Social Sciences Communications*, 9(1), 1-15.

Drennan, L., McConnell, A. and Stark, A. (2015). *Risk and crisis management in the public sector* (2nd ed.). Routledge.

Druckman, J.N. (2017). The crisis of politicization within and beyond science. *Nature Human Behaviour*, 1(9), 615-617.

Easton, M., De Paepe, J., Evans, P., Head, B.W., and Yarnold, J. (2022). Embedding expertise for policy responses to COVID-19: comparing decision-making structures in two federal democracies. *Public Organization Review*, 22(2), 309-326.

Esguerra, A., Beck, S. and Lidskog, R. (2017) 'Stakeholder engagement in the making: IPBES legitimization politics', *Global Environmental Politics*, 17(1), 59-76.

Eyal. G. (2019). *The crisis of expertise*. Cambridge: Polity.

Fischer, F. (1990). *Technocracy and the politics of expertise*. Sage.

Fischer, A., Wentholt, M., Rowe, G., and Frewer, L. (2014). Expert involvement in policy development: a systematic review of current practice. *Science and Public Policy*, 41(3), 332-343.

Frewer, L., and Salter, B. (2002). Public attitudes, scientific advice and the politics of regulatory policy: the case of BSE, *Science and Public Policy*, 29(2), 137-145.

Galanti, M. and Saracino, B. (2021). Inside the Italian COVID-19 task forces. *Contemporary Italian Politics*, 13(2), 275-291.

Gieryn, T.F. (1983). Boundary work and the demarcation of science from non-science, *American Sociological Review*, 48(6), 781-795.

Goldman, A.I. (2001). Experts: which ones should you trust? *Philosophy and Phenomenological Research*, 63(1), 85-110.

Groux, G., Hoffman, S. and Ottersen, T. (2018). A typology of scientific advisory committees. *Global Challenges*, 2(9), 1800004.

Grundmann, R. (2018). The rightful place of expertise, *Social Epistemology*, 32(6), 372-386.

Gullberg, A. T. (2008b). Lobbying friends and foes in climate policy: The case of business and environmental interest groups in the European Union. *Energy Policy*, 36(8), 2964-2972.

Hadorn, S., Sager, F., Mavrot, C., Malandrino, A., and Ege, J. (2022) Evidence-based policymaking in times of acute crisis: Comparing the use of scientific knowledge in Germany, Switzerland, and Italy. *Politische Vierteljahresschrift*, 63(2), 359-382.

Head, B.W. (2016). Toward more ‘evidence-informed’ policy making? *Public Administration Review*, 76(3), 472-484.

Hodges, R., Caperchione, E., van Helden, J., Reichard, C., and Sorrentino, D. (2022). The role of scientific expertise in COVID-19 policy-making: evidence from four European countries. *Public Organization Review*, 22(2), 249-267.

Hordijk, M., and Baud, I. (2006). The role of research and knowledge generation in collective action and urban governance: How can researchers act as catalysts? *Habitat International*, 30(6), 668-689.

Howarth, C., Parsons, L. and Thew, H. (2020). ‘Effectively communicating climate science beyond academia‘, *One Earth*, 2(4), 320-324.

Howlett, M., Capano, G. and Ramesh, M. (2018). Designing for robustness: surprise, agility and improvisation in policy design, *Policy & Society*, 37(4), 405-421.

Hustedt, T., and Veit, S. (2017). Policy advisory systems: change dynamics and sources of variation. *Policy Sciences*, 50(1), 41-46.

Jensen, M.D., Lynggaard, K. and Kluth, M. (2022). Paths, punctuations, and policy learning: comparing patterns of European use of scientific expertise during the COVID-19 crisis, *Public Organization Review*, 22(2), 223-247.

Kingdon, J. (1995). *Agendas, alternatives and public policies*, 2nd edition. London: Pearson.

Lacey, J., Howden, M., Cvitanovic, C., and Colvin, R. (2018). Understanding and managing trust at the climate science–policy interface. *Nature Climate Change*, 8(1), 22-28.

Laage-Thomsen, J., and Frandsen, S. (2022). Pandemic preparedness systems and diverging COVID-19 responses ….. expert perceptions of pandemic response in Denmark, Norway, and Sweden. *Globalization and Health*, 18(3).

Laureys, V., and Easton, M. (2019). Resilience of public and private security providers: a state-of-the-art literature review. *Policing – An International Journal* 42(2), 126-140.

Lavazza, A. and Farina, M. (2020). The role of experts in the COVID-19 pandemic and the limits of their epistemic authority in democracy, *Frontiers in Public Health*, 8, art356.

Liberatore, A., and Funtowicz, S. (2003). 'Democratising' expertise, 'expertising' democracy: What does this mean, and why bother? *Science and Public Policy*, 30(3), 146-150.

Livingston, J. (2023). A global goal for climate change: the place of science in politics. *Social Research*, 90(1), 111-126.

Lynggaard, K., Jensen, M., and Kluth, M. (eds) (2023). *Governments' responses to the Covid-19 pandemic in Europe*. Cham: Palgrave.

MacAulay, M., Fafard, P., Cassola, A. and Palkovits, M. (2023) Analysing the 'follow the science' rhetoric of government responses to COVID-19, *Policy & Politics*, XX(XX): 1–20.

Majcen, Š. (2017). Evidence based policy making in the European Union: the role of the scientific community. *Environmental Science and Pollution Research*, 24(9), 7869-7871.

Mergel, I., Gong, Y., Bertot, J. (2018) Agile government: Systematic literature review and future research. *Government Information Quarterly*, 35(2), 291-298.

Moore, A. and MacKenzie, M. (2020) Policy making during crises: how diversity and disagreement can help manage the politics of expert advice. *British Medical Journal*, 371 (4039), 1-3.

Nichols, T. (2017) *The death of expertise*. Oxford: Oxford University Press.

OECD (2015). Scientific advice for policy making: the role and responsibility of expert bodies and individual scientists. Paris: OECD STI Policy Paper 21.

Pattyn, V., Matthys, J. and van Hecke, S. (2021). High-stakes crisis management in the Low Countries: comparing government responses to COVID-19, *International Review of Administrative Sciences*, 87(3): 593–611.

Page, E.C. (1999). The insider/outsider distinction. *British Journal of Politics and International Relations,* 1(2), 205-214.

Page, E.C. (2010). Bureaucrats and expertise. *Sociologie du travail*, 52 (2), 255-273.

Petridou, E., and Mintrom, M. (2021). A research agenda for the study of policy entrepreneurs. *Policy Studies Journal*, 49(4), 943-967.

Pielke, R. A. (2007) *The honest broker: Making sense of science in policy and politics*, Cambridge University Press.

Poulsen, B. (2022). When civil servants go frontstage – the mediatization of the role of the civil servant during the COVID-19 crisis. *Administrative Sciences*, 12, 73. https://doi.org/10.3390/admsci12030073

Pralle, S. B. (2009). Agenda-setting and climate change, *Environmental Politics*, 18(5), 781- 799.

Radaelli, C. (1999). The public policy of the European Union: whither politics of expertise? *Journal of European Public Policy*, 6(5), 757-774.

Rajan, D., Koch, K., et al. (2020) Governance of the COVID-19 response: A call for more inclusive and transparent decision- making. *BMJ Global Health*, 5(5), 1-8.

Rubin, O., Errett, N., Upshur, R., and Baekkeskov, E. (2021). The challenges facing evidence-based decision making in the initial response to COVID-19. *Scandinavian Journal of Public Health*, 49(7), 790-796.

Staupe-Delgado, R. and Rubin, O. (2022). Challenges associated with creeping disasters in disaster risk science and practice: considering disaster onset dynamics, *International Journal of Disaster Risk Science*, 13(1), 1-13.

Topp, L., Mair, D., Smillie, L., and Cairney, P. (2018). Knowledge management for policy impact: the case of the European Commission's Joint Research Centre, *Palgrave Communications*, 4(87), 1-10.

Weber, M. (1958). *From Max Weber: essays in sociology*, eds H.Gerth and C.W. Mills. Oxford University Press.

Weible, C. (2008). Expert-based information and policy subsystems: a review and synthesis. *Policy Studies Journal*, 36(4), 615–635.

Weible, C., Pattison, A. and Sabatier, P. (2010). Harnessing expert-based information for learning and the sustainable management of complex socio-ecological systems, *Environmental Science & Policy*, 13(6), 522-534.

Weible, C., Ingold, K., Nohrstedt, D., Henry, A. and Jenkins-Smith, H. (2020). Sharpening advocacy coalitions. *Policy Studies Journal*, 48(4), 1054-1081.

Weingart, P. (1999). Scientific expertise and political accountability: Paradoxes of science in politics, *Science and Public Policy*, 26(3), 151-161.

Witter, S., Anderson, I., et al. (2019). What, why and how do health systems learn from one another? Insights from eight low-and middle-income country case studies, *Health Research Policy and Systems*, 17(1), 1-13.

Yakubovich, V., and Burg, R. (2019). Friendship by assignment? From formal interdependence to informal relations in organizations. *Human Relations*, 72(6), 1013-1038.

Zaki, B.L., and Wayenberg, E. 2020. Shopping in the scientific marketplace: COVID-19 through a policy learning lens. *Policy Design and Practice*, 4(1), 15-32.

Title	**Issue**	**Type of study**	**Crisis focus**	**Authors**
Expert perspectives on the changing dynamics of policy advisory systems.	Focus on expert experiences in Belgium and Australia	Qualitative, case study, comparative	COVID	Easton, Yarnold, Vervaenen, De Paepe & Head
How data-sharing ecosystems respond to compounding crises	Data ecosystems in the same case in Australia	Qualitative, case study	COVID	Howard & Hyland-Wood
Between Naked power and Technocracy	Politicization in three case studies – Denmark, Greece, and the US	Qualitative, case study, comparative	COVID	Aagaard, Poulsen & Chatzopoulou
Between Depoliticization and Politicization: the role of expertise in Covid-19 politics in Europe	Politicization in 31 EU countries	Quantitative, comparative	COVID	Lynggaard, Exadaktylos, Jensen & Kluth
Evidence-to-policy pathways and pandemic governance	Comparative institutional insights from COVID-19 responses in six LMIC countries (Panama, Jamaica, Sri Lanka, Indonesia, Kenya, and Zimbabwe)	Qualitative, comparative	COVID	Naomi, Allen, Rima Mbaabu, Mendes, Ndongwe, Djindan, Gardia de Paredes & Kandanamull
The curse of complacency – the scope for expert-led agenda setting during creeping crises	Creeping crisis– entering climate debate.	Qualitative, case study, comparative	Climate crisis	Zaman, Rubin & Delgado
The evolution of knowledge exchange architecture for climate action, with politicisation of climate change in the European Union	EU climate policy & knowledge architecture.	Qualitative, historical	Climate crisis	Dupont, Rosamond & Zaki

Table 1: Overview of contributions in this special issue.